

A Neural Dirichlet-Neumann Approach for Learning Euler Equation Flows

Jonathan Ma
Department of Mathematics
University of Delaware
johnma@udel.edu

Samuel Zheng
Affiliation
Address
email

Coauthor
Affiliation
Address
email

Ke Chen
Department of Mathematics
University of Delaware
kechen@udel.edu

Abstract

Abstract to be completed.

Keywords: Water waves; Dirichlet–Neumann operator; neural operators

Working visualization draft. The figures use the historical C27 checkpoint and v9 data, together with archived guarded rollouts. They are not a new comprehensive, trajectory-disjoint evaluation. A matched vanilla-FNO comparison is not yet available. Timing measurements are CPU-only. The method sections below retain the supplied manuscript’s unfinished derivations and author notes.

1 Introduction

More needs to be added to introduction.

Governing Equations & Hamiltonian Formulation Obviously we should refine this portion later. We consider the evolution of a free surface on top of a two-dimensional fluid under the influence of gravity. We let x denote our horizontal direction, and y denote our vertical direction, and t denote time. We assume periodic boundary conditions on the horizontal, with domain length $L > 0$. Let $y = \eta(x, t)$ represent our free surface. We fix an impermeable bottom at a constant depth $y = -h$. We assume the fluid is incompressible, inviscid, and that its flow is irrotational, so that the fluid velocity is given by $u = \nabla\varphi$, and its velocity potential φ satisfies Laplace’s equation:

$$\Delta\varphi = 0, \tag{1}$$

in the fluid region bounded below by the bottom at $y = -h$, and above by the free surface defined by $\eta(x, t)$. We assume the boundary conditions

$$\begin{aligned} \varphi_y &= 0, & \text{on } y = -h, \\ \eta_t + \nabla_x \eta \cdot \nabla_x \varphi - \varphi_y &= 0, & \text{on } y = \eta, \\ \varphi_t + \frac{1}{2}|\nabla\varphi|^2 + g\eta &= 0, & \text{on } y = \eta, \end{aligned} \tag{2}$$

where g denotes the acceleration due to gravity. We reduce the dimensionality of the classical potential flow formulation of the Euler equation problem by introducing the Dirichlet–Neumann operator (DNO), which

takes Dirichlet data $\xi(x, t) = \varphi(\eta(x, t), t)$ and gives the corresponding Neumann data, i.e., the normal velocity at the free surface. The DNO G satisfies

$$G(\eta)\xi = (-\nabla_x \eta, 1)^\top \cdot \nabla \varphi|_{y=\eta}.$$

The system is Hamiltonian in η, ξ , with

$$H(\eta, \xi) = \frac{1}{2} \int_{\Omega} [\xi G(\eta)\xi + g\eta^2], dx. \quad (3)$$

Using the fact that

$$\eta_t = \frac{\delta H}{\delta \xi}, \quad \xi_t = -\frac{\delta H}{\delta \eta}, \quad (4)$$

we can write the system (2) as

$$\begin{aligned} \eta_t &= G(\eta)\xi, \\ \xi_t &= -g\eta - \frac{1}{2}\xi_x^2 + \frac{(G(\eta)\xi + \eta_x \xi_x)^2}{2(1 + \eta_x^2)}. \end{aligned} \quad (5)$$

Throughout this manuscript, we note that the operator $G(\eta)$ implicitly depends on h . When it is noteworthy, we note this dependence by writing the DNO as $G(\eta; h)$. For each admissible surface η , the operator $G(\eta)$ is linear in its Dirichlet argument ξ , self-adjoint, and positive semidefinite; it is positive definite after restriction to an appropriate mean-zero space. The Hamiltonian H of our system is invariant to time. In addition to being vital as a stability check for our classical algorithms, we will make use of energy conservation to increase the stability of our neural network models.

1.1 Related Work

1.2 Contribution

The primary objective of this work is to construct a neural approximation of the water-wave DNO whose evaluation is independent of the Craig–Sulem truncation order M , while retaining properties needed for reliable time integration. In particular, the proposed method is designed to achieve accurate DNO evaluation across several challenging solution classes, approximate Hamiltonian conservation, structural consistency, and stable long-time trajectories. The intended principal comparison is with a vanilla FNO trained only with the data-inconsistency loss, for which small one-step prediction errors need not imply structure-preserving or stable dynamics. [This matched baseline comparison remains to be completed; the present figures show C27 and Craig–Sulem results only.](#) Computational cost is examined as a practical consequence of the M -independent representation, rather than as the principal contribution.

2 Problem Setup

We now describe methods for solving the water wave equations in terms of solving for the DNO.

2.1 Forward Problem Solver

Craig–Sulem Series [Add citations later.](#) It is known that if η is Lipschitz, then G is analytic in η , hence G has a convergent Taylor series. We write

$$G(\eta) = \sum_{i \geq 0} G_i(\eta), \quad (6)$$

where G_i is the i th homogeneous term in the Taylor expansion of G . It can be shown that each term G_i can be calculated using an explicit recursion formula with base case $G_0 = |D_x| \tanh(h|D_x|)$, where $D_x = -i\nabla_x$, and i is the imaginary unit. We note that G_0 is thus independent of the surface shape η . By use of dynamic programming and the fast Fourier transform, it can be shown that the time complexity of calculating the M th polynomial G_M is $\Theta(M^2 N \log N)$, where N is the spatial discretization dimension. Xu and Guyenne make use of this recursive method of calculating G up to the M th Taylor term in [\[cite\]](#) to solve the water wave problem. The details of this algorithm are in the appendix.

Numerical Methods for CS Series Add typical method for calculating the DNO. Add challenges of calculating DNO.

Time Integrator Following Xu and Guyenne, we employ an implicit symplectic scheme to solve our time stepping PDE problem. Details in appendix. Sam

2.2 Learning Problem

Describe learning the DNO operator using a parametrized model, and a dataset $\mathcal{D}$ and loss $\mathcal{L}$. Need to emphasize that the target operator is depth dependent and the PDE is time dependent. Want to learn many classes of initial conditions and maintain long-time stability.

Deep Learning / Scientific Machine Learning Deep learning methods have recently seen success in being incorporated into solutions for scientific problems, especially those governed by partial differential equations. This has given rise to the field of Scientific Machine Learning. Practitioners leverage the well-known universal approximation results of deep neural networks to model unknown target functions using data. When extended to the appropriate domains and function spaces of partial differential equation solutions, this is often referred to as operator learning. Given the goal of replacing the costly computation of the DNO with a cheap-to-evaluate and accurate neural surrogate, we require describing a parametric neural network model G_θ , a dataset $\mathcal{D}$ to learn from, and a loss function $\mathcal{L}$ to enforce consistency of our parametric model with the target operator G . We describe our parametric family of models in Section 3.5, the construction of $\mathcal{D}$ in Section 4.1, and our choice of $\mathcal{L}$ in Section 3.1.

Operator Learning Operator learning is broadly characterized by methods for approximating mappings between Hilbert or Banach spaces. In practice, this is achieved by designing methods of approximation which are invariant to choice of discretization of the domain Ω . We achieve this by using a Fourier Neural Operator (FNO) backbone.

3 Proposed Method

We propose a method for solving the water wave equations by approximating the DNO. Additionally, we demonstrate that simply approximating the DNO pointwise is insufficient for maintaining stability in the downstream numerical PDE task. Thus we provide several physics-informed methods for ensuring that the learned DNO provides the necessary stability for PDE simulation.

3.1 Designing a Loss Function

We teach our model the law of the DNO operator by penalizing the following loss:

$$\mathcal{L}_{\text{DNO}} = \alpha_1 \mathcal{L}_{L^2} + \alpha_2 \mathcal{L}_{\text{MB}} + \alpha_3 \mathcal{L}_{\text{TG}} + \alpha_4 \mathcal{L}_{\text{HD}}, \quad (7)$$

where the coefficients α_i are a convex combination determined through hyperparameter search. The loss $\mathcal{L}_{L^2}$ is the ordinary L^2 loss, calculated spectrally. **KC: New notations should be defined.** The last three terms of our aggregate loss are each derived from the Hamiltonian of our system, helping ensure the learned dynamics of our network are consistent with physics. Each loss is carefully ablated.

3.2 Hamiltonian Mode-Balancing Penalty

Need to numerically show that during water wave simulation it is possible that certain modes tend to zero or certain modes completely dominate the output. As we are learning our operator chiefly in Fourier space, and using a spectral solver in the downstream PDE simulation, we aim to ensure that our learned operator respects the evolution of each active Fourier mode in a given instantiation of our time-stepping PDE. A key challenge in ensuring this is disallowing certain modes to dominate the evolution. Add supporting references. We show that a single-step loss which penalizes long-term mismatches in Fourier mode activity can be derived

from the Hamiltonian of our problem. We begin by deriving a quadratic approximation to our Hamiltonian (3). Define $F(\varepsilon) := G(\varepsilon\eta)\xi$, for fixed η, ξ , and $\varepsilon > 0$. Applying a Taylor expansion on F about zero gives

$$F(\varepsilon) = F(0) + \varepsilon F'(0) + \mathcal{O}(\varepsilon^2). \quad (8)$$

Perturbing (3) by ε ,

$$H(\varepsilon\eta, \varepsilon\xi) = \frac{1}{2} \int_{\Omega} \varepsilon\xi \cdot G(\varepsilon\eta)\varepsilon\xi \, dx + \frac{g}{2} \int_{\Omega} \varepsilon^2\eta^2 \, dx \quad (9)$$

$$= \frac{1}{2} \int_{\Omega} \varepsilon^2\xi[F(0) + \varepsilon F'(0) + \mathcal{O}(\varepsilon^2)]\xi \, dx + \frac{g}{2} \int_{\Omega} \varepsilon^2\eta^2 \, dx \quad (10)$$

$$= \varepsilon^2 \left(\frac{1}{2} \int_{\Omega} \xi F(0) + \frac{g}{2} \int_{\Omega} \eta^2 \, dx \right) + \varepsilon^3 \left(\frac{1}{2} \int_{\Omega} F'(0) \right) + \mathcal{O}(\varepsilon^4). \quad (11)$$

Therefore the quadratic Hamiltonian is

$$H_2(\eta, \xi) = \frac{1}{2} \int_{\Omega} [\xi G(0)\xi + g\eta^2] \, dx. \quad (12)$$

The Fourier symbol of the flat-surface DNO, $G(0)$, has known form:

$$\widehat{G(0)}\xi_k = |k| \tanh(h|k|)\widehat{\xi}_k. \quad (13)$$

Thus by using Parseval's theorem, one may calculate that for any η, ξ , $H_2(\eta, \xi) = \sum_k H_2(\eta, \xi)_k$, where

$$H_2(\eta, \xi)_k = \frac{1}{2} \left(|k| \tanh(h|k|) |\widehat{\xi}_k|^2 + g |\widehat{\eta}_k|^2 \right). \quad (14)$$

Plugging (12) into (4) yields the linearized system

$$\eta_t^L = G(0)\xi, \quad \xi_t^L = -g\eta. \quad (15)$$

Taking the Fourier transform yields, for nonzero modes k ,

$$\partial_t \widehat{\eta}_k^L = \widehat{G(0)}\xi_k, \quad \partial_t \widehat{\xi}_k^L = -g\widehat{\eta}_k. \quad (16)$$

Differentiating the first equation once more in time,

$$\partial_{tt} \widehat{\eta}_k^L = \widehat{G(0)}_k \partial_t \widehat{\xi}_k \quad (17)$$

$$= -g \widehat{G(0)}_k \widehat{\eta}_k^L. \quad (18)$$

Define $\omega_k^2 := g \widehat{G(0)}_k$, then note that

$$\partial_{tt} \widehat{\eta}_k^L + \omega_k^2 \widehat{\eta}_k^L = 0 \quad (19)$$

is a harmonic oscillator with frequency ω_k , whose amplitude is

$$A_k^2 = |\widehat{\eta}_k^L|^2 + \frac{|\partial_t \widehat{\eta}_k^L|^2}{\omega_k^2}. \quad (20)$$

Furthermore, this quantity is invariant to time:

$$\begin{aligned} \frac{d}{dt} A_k^2 &= \partial_t \widehat{\eta}_k^L \overline{\widehat{\eta}_k^L} + \widehat{\eta}_k^L \partial_t \overline{\widehat{\eta}_k^L} \\ &\quad + \frac{1}{\omega_k^2} \left(\partial_{tt} \widehat{\eta}_k^L \partial_t \overline{\widehat{\eta}_k^L} + \partial_t \widehat{\eta}_k^L \partial_{tt} \overline{\widehat{\eta}_k^L} \right) \\ &= \partial_t \widehat{\eta}_k^L \overline{\widehat{\eta}_k^L} + \widehat{\eta}_k^L \partial_t \overline{\widehat{\eta}_k^L} \\ &\quad + \frac{1}{\omega_k^2} \left(\omega_k^2 \widehat{\eta}_k^L \partial_t \overline{\widehat{\eta}_k^L} + \omega_k^2 \partial_t \widehat{\eta}_k^L \overline{\widehat{\eta}_k^L} \right) \\ &= 0. \end{aligned}$$

We use this quantity to quantify the scale of the k th Fourier mode of the target $G(\eta)\xi$, for fixed η, ξ .

3.3 Tangent-Gain Penalty

For fixed η, ξ , write the error of our model G_θ against G by $\mathcal{E} = G_\theta(\eta)\xi - G(\eta)\xi$. If we assume that for some $c > 0$ our wave has a transporting profile, $G(\eta)\xi := -c\eta_x$, then, assuming $G_\theta(\eta)\xi \approx (1 + \varepsilon)G(\eta)\xi$, we find $\mathcal{E} \approx -\varepsilon c\eta_x$, which is an accelerative error in the translating direction. We empirically find that when simulating the water wave equations using a naïve neural surrogate with certain initial wave profiles, translating error dominates the error accumulated. [Add a small numerical experiment to support this](#). We thus provide a simple loss which targets general potential translation error accumulated during the simulation of the water wave PDE. For fixed x, η , and $a > 0$, the velocity of the translating operator $T_a(\eta) := \eta(x - a)$ is $-\eta_x$. Let P_{η_x} be the L^2 projection operator onto $\text{span}\{\eta_x\}$. As before, let $\mathcal{E} = G_\theta(\eta)\xi - G(\eta)\xi$. We would like to penalize

$$\frac{\|P_{\eta_x}\mathcal{E}\|_2^2}{\|\eta\|_2^2} = \frac{|\langle \mathcal{E}, \eta_x \rangle|^2}{\|\eta\|_2^2 \|\eta_x\|_2^2}, \quad (21)$$

where we divide by the square norm of η as is practice in deep learning to maintain scale invariance of our loss function. In practice, a target wave profile may have more than one translating component, with varying speeds, which motivates deriving a localized version of (21). We first define the periodic Gaussian kernel

$$K_\sigma(x - y) := \frac{1}{\sqrt{2\pi\sigma^2}} \sum_{m \in \mathbb{Z}} \exp\left(-\frac{(x - y + mL)^2}{2\sigma^2}\right),$$

for $x, y \in \mathbb{T}_L$ and $\sigma^2 > 0$ the variance/localization parameter of the kernel. We then define the following averaging operator:

$$(B_\sigma f)(x) := \int_0^L K_\sigma(x - y) f(y) dy.$$

For $k_n = 2\pi n/L$, the n th Fourier series coefficient of our kernel is $\widehat{K}_{\sigma,n} = L^{-1} \exp(-\sigma^2 k_n^2/2)$. For a periodic function $f(x) = \sum_{n \in \mathbb{Z}} \widehat{f}_n \exp(ik_n x)$, by the convolution theorem,

$$(\widehat{B_\sigma f})_n = \exp(-\sigma^2 k_n^2/2) \widehat{f}_n.$$

The above Fourier multiplier form of B_σ is useful for us as numerically we calculate this convolution spectrally rather than physically. Upon defining our averaging operator, we replace our L^2 inner product with our localized inner product $\langle f, g \rangle_{\sigma,x} = \int_{\mathbb{T}_L} K_\sigma(x, y) f(y) g(y) dy$. We repeat the above step of projecting the error $\mathcal{E}$ onto the span of η_x under our new weighted inner product using the projection operator $P_{\eta_x}^{\sigma,x_0}$, where we emphasize the dependence on the fixed reference point x_0 :

$$\frac{\|P_{\eta_x}^{\sigma,x_0} \mathcal{E}\|_{\sigma,x_0}^2}{\|\eta\|_2^2} = \frac{|\langle \mathcal{E}, \eta_x \rangle_{\sigma,x_0}|^2}{\|\eta\|_2^2 \|\eta_x\|_{\sigma,x_0}^2}.$$

Lastly, we integrate over all reference points x_0 to define our loss $\mathcal{L}_{\text{TG}}$:

$$\mathcal{L}_{\text{TG}} = \frac{1}{\|\eta\|_2^2} \int_{\mathbb{T}_L} \|P_{\eta_x}^{\sigma,x_0} \mathcal{E}\|_{\sigma,x_0}^2 dx_0. \quad (22)$$

3.4 Hadamard Derivative

[Jonathan](#) Intuitively learning the DNO pointwise from data is insufficient to guarantee stability during simulation of the time-stepping PDE. Notably, as our network only claims to be an approximation to the true DNO, even small numerical error in its approximation will force the network to make predictions on future states which may only necessarily be near the true PDE trajectory. We cite the following theorem of Lannes which can help encourage our network to be locally accurate in an appropriate manner.

Proposition 1. *Let η, ξ , and ζ be sufficiently smooth, and assume that the fluid depth remains strictly positive. Define*

$$q = G(\eta)\xi, \quad B = \frac{q + \eta_x \xi_x}{1 + \eta_x^2}, \quad V = \xi_x - \eta_x B.$$

Then the map $\eta \mapsto G(\eta)\xi$ is Fréchet differentiable and

$$D_\eta G(\eta)[\zeta]\xi = -G(\eta)(\zeta B) - \partial_x(\zeta V).$$

A proof is provided in the appendix, as well as [citations](#). To enforce local correctness of our learned model G_θ , we define the residual of G_θ with Proposition 1:

$$\mathcal{H}_\theta(\eta, \xi; \zeta) = D_\eta G_\theta(\eta)[\zeta]\xi + G_\theta(\eta)(\zeta B_\theta) + \partial_x(\zeta V_\theta),$$

where

$$B_\theta = \frac{G_\theta(\eta)\xi + \eta_x \xi_x}{1 + \eta_x^2}, \quad V_\theta = \xi_x - \eta_x B_\theta.$$

We approximate this Fréchet derivative by sampling a random direction ζ on the unit sphere S^{d-1} , where d is our discretization dimension, and sampling $\varepsilon \sim \mathcal{U}(10^{-3}, 3 \times 10^{-3})$, then we calculate

$$D_\eta G_\theta(\eta)[\zeta]\xi \approx \frac{G_\theta(\eta + \varepsilon \zeta)\xi - G_\theta(\eta)\xi}{\varepsilon}.$$

The terms $\partial_x(\zeta V_\theta)$, η_x , and ξ_x are calculated by spectral differentiation. We then define the loss $\mathcal{L}_{\text{HD}}$ as

$$\mathcal{L}_{\text{HD}} = \mathbb{E}_{\substack{\zeta \sim \mathcal{U}(S^{d-1}) \\ \varepsilon \sim \mathcal{U}(10^{-3}, 3 \times 10^{-3})}} [\mathcal{H}_\theta(\eta, \xi; \zeta)]. \quad (23)$$

3.5 Architecture

We now describe the architectural decisions we made in defining our family of parametric models for learning the DNO. Our network is built chiefly in Fourier space, as a composition of learned Fourier multipliers. We show that this allows our network to satisfy several properties of the DNO by construction, as well as enabling the discretization invariance trait which is desirable in operator learning.

CS Truncation The first term of (6) has a known Fourier symbol:

$$\widehat{G_0 \xi_k} = |k| \tanh(h|k|) \widehat{\xi_k}. \quad (24)$$

The second term of (6), through the known Craig–Sulem recursion, is written

$$G_1(\eta)\xi = D \cdot (\eta D\xi) - G_0(\eta G_0 \xi). \quad (25)$$

We thus propose to learn the following approximation:

$$G(\eta)\xi \approx G_0 \xi + G_1(\eta)\xi + G_\theta(\eta)\xi, \quad (26)$$

where G_θ is our learned surrogate. [We should quantify the actual error between \$G\$ and \$G_0 + G_1\$ to give some motivation. It is of course \$\mathcal{O}\(\eta^2\)\$.](#)

Low Rankedness of Residual [Jonathan](#)

Depth and Wavenumber Conditioning As mentioned in Section 1, the DNO depends on the geometry of the prescribed water wave problem. In our two-dimensional fluid setting, this geometry dependence is explicit in the depth of our impermeable bottom at depth $-h$. In addition, due to the depth dependence in the base case of our CS recursion (24), the depth parameter h acts mode-by-mode on the DNO. [Jonathan](#)

Positive Definiteness and Linearity As mentioned in Section 1, for each admissible η the target operator $G(\eta)$ is linear in ξ and positive semidefinite, which motivates the following architectural decisions for the neural surrogate G_θ .

3.6 Training the Neural Operator

Jonathan

4 Numerical Experiments

In this section, we illustrate the proposed numerical study using the historical C27 checkpoint trained on the v9 snapshot archive. The available figures evaluate DNO approximation, structural properties, trajectory accuracy, spectral behavior, Hamiltonian drift, and computational cost. For this section, $G_\theta(\eta; h)\xi$ denotes the complete deployed surrogate, including its analytic Craig–Sulem terms. The intended comparison with a vanilla FNO trained only using the data-inconsistency loss remains pending; no unmatched model is substituted for that baseline. The results below are preliminary illustrations of the available C27 evaluation, not the final comprehensive testing protocol.

4.1 Numerical Setup and Data Generation

4.1.1 Wave Regimes

The intended numerical study spans traveling, interacting, broadband, and modulational dynamics. Its four principal solution classes are:

1. Stokes traveling waves;
2. Tanaka solitary waves, including isolated and multiple-wave cases;
3. JONSWAP/TMA irregular waves; and
4. Benjamin–Feir configurations with a carrier and perturbing sidebands.

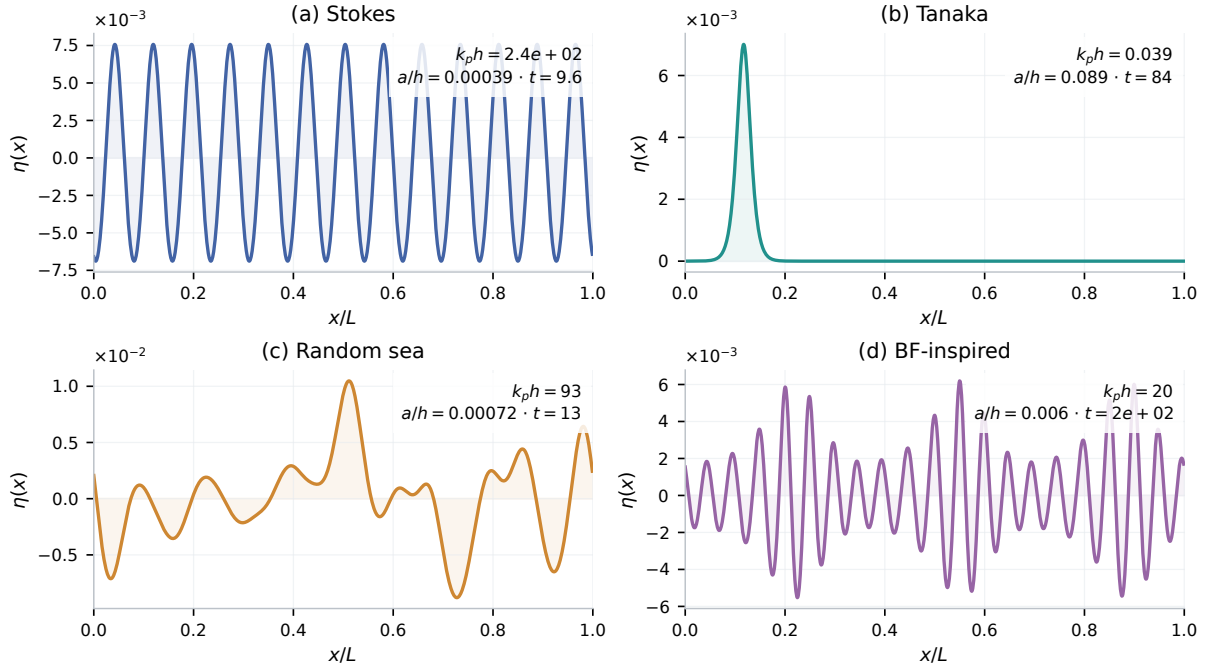
The historical v9 data used for the present figures contain Stokes, Tanaka, random-sea, and BF-inspired families. We retain these historical names: the random-sea and BF-inspired samples are not assumed to coincide with the newer JONSWAP/TMA and canonical Benjamin–Feir constructions. Additional linear and manufactured sources in v9 are outside the four-family plots. [The final dataset table should give the parameter ranges, numbers of independent initial conditions, retained snapshots, and split counts for the finalized data-generation protocol.](#)

4.1.2 Reference Solver and Dataset Construction

The available historical trajectories use a periodic domain of length $L = 2\pi$, $N_x = 1024$ grid points, and gravity $g = 1$. The reference DNO is computed with Craig–Sulem order $M_{\text{ref}} = 6$ and padding factor eight. Reference and learned trajectories use the same integrating-factor Gauss–Legendre scheme, with four fixed-point iterations, time step $\Delta t = 0.01$, and a hard spectral cutoff at mode 128. The C27 archives also use a soliton-selective spectral guard; the plotted trajectory results are conditional on that guarded deployment protocol.

For one-step DNO evaluation we sample the original v9 snapshot archive and reconstruct its seed-zero 80/10/10 training, validation, and test row split. We evaluate 512 training rows and 512 held-out rows independently within each of the four main families, for 4096 samples total. This historical row split is not a trajectory-disjoint split, and nearby snapshots from the same underlying trajectory may occur in different subsets.

The trajectory figures reuse nine historical evaluation panels, with 32 attempted cases per panel. Three reference solves are invalid, leaving 285 paired cases: 63 Stokes, 64 Tanaka, 64 random-sea, and 94 BF-inspired. These source-family panels are not asserted to be held out from the original training trajectories. The integration horizons are $T = 20$ for Stokes and random-sea cases and $T = 200$ for Tanaka and BF-inspired cases. [A final generalization study should instead use complete, independent initial-condition trajectories for each train/validation/test split and report the rejection and subsampling rules explicitly.](#)



Actual v9 snapshots · k_p from the surface spectrum · $a = (\max \eta - \min \eta)/2$

Figure 1: Actual v9 training snapshots from the four historical families. One sample is selected at the median sampled C27 DNO error within each family. The dominant nonzero surface Fourier mode defines k_p , and $a = (\max \eta - \min \eta)/2$. Each panel reports $k_p h$, a/h , and the saved time. The random-sea and BF-inspired labels refer to the old v9 generators, not the finalized JONSWAP/TMA or Benjamin–Feir definitions.

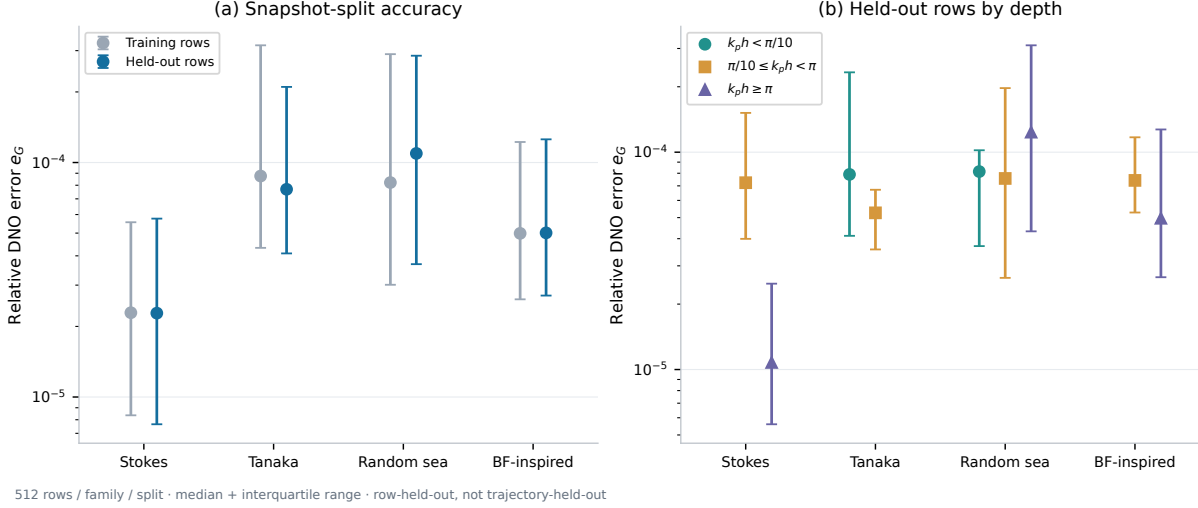


Figure 2: Relative L^2 DNO errors on sampled v9 rows. Left: training and held-out medians with interquartile bars, using 512 rows per family per split. Right: held-out rows grouped by the stated depth bins. Missing groups contain no sampled rows. The original snapshot split is used; these results do not establish trajectory-disjoint generalization.

4.1.3 Implementation Details

The figures use the final, epoch-40 checkpoint from the July C27 run and its saved normalization. Fresh model evaluations and timing measurements use double precision. Historical trajectory arrays were stored in single precision and converted to double precision for the present diagnostics. The timing experiments run on an AMD EPYC 9124 CPU host with an affinity of eight logical CPUs; they are not GPU measurements. The complete training configuration, loss weights, checkpoint-selection rule, and number of independent training runs remain to be added to the final implementation paragraph. No new training was performed for this visualization draft.

4.2 Accuracy and Structural Properties of the Learned DNO

4.2.1 Relative DNO Accuracy

For sample j , define

$$e_G^{(j)} = \frac{\|G_\theta(\eta^{(j)}; h^{(j)})\xi^{(j)} - G(\eta^{(j)}; h^{(j)})\xi^{(j)}\|_{L^2}}{\|G(\eta^{(j)}; h^{(j)})\xi^{(j)}\|_{L^2}}. \quad (27)$$

In the present snapshot experiment, the reference values are the stored v9 DNO targets. Figure 2 reports classwise medians and interquartile ranges and resolves held-out rows by the depth parameter $k_p h$. We use the explicit bins $k_p h < \pi/10$, $\pi/10 \leq k_p h < \pi$, and $k_p h \geq \pi$, with k_p the strongest nonzero Fourier mode of the sampled surface. The sampled held-out medians are 2.28×10^{-5} for Stokes, 7.69×10^{-5} for Tanaka, 1.09×10^{-4} for random sea, and 5.01×10^{-5} for BF-inspired data. These are sampled row-split statistics, not full-dataset or trajectory-disjoint errors.

4.2.2 Linearity and Positive Semidefiniteness

For fixed (η, h) , Dirichlet data ξ_1, ξ_2 , and scalars a, b , define

$$e_{\text{lin}} = \frac{\|G_\theta(\eta; h)(a\xi_1 + b\xi_2) - aG_\theta(\eta; h)\xi_1 - bG_\theta(\eta; h)\xi_2\|_{L^2}}{|a|\|G_\theta(\eta; h)\xi_1\|_{L^2} + |b|\|G_\theta(\eta; h)\xi_2\|_{L^2}}. \quad (28)$$

We perform 100 independent trials per family. The surface-depth pair and two Dirichlet inputs are drawn independently from training snapshots of that family; the inputs are centered, and a, b are sampled independently from $\mathcal{U}(-1, 1)$. Mean linearity residuals range from 1.08×10^{-14} to 4.79×10^{-13} across the four families.

Because constant Dirichlet data belong to the DNO nullspace, its quadratic form is positive semidefinite on the full space. For $\tilde{\xi} = \xi - L^{-1} \int_0^L \xi(x) dx$, define

$$q_\theta(\eta, \tilde{\xi}; h) = \frac{\langle \tilde{\xi}, G_\theta(\eta; h) \tilde{\xi} \rangle_{L^2}}{\|\tilde{\xi}\|_{L^2}^2}. \quad (29)$$

On 100 independently sampled matched training snapshots per family, no quadratic form falls below $-\tau$, with $\tau = 10^{-10}$. This finite sample test is not a proof of positive semidefiniteness of the learned operator. [Add the matched vanilla-FNO structural tests and the final side-by-side table.](#)

4.3 Trajectory Accuracy and Long-Time Stability

For either learned model in the intended comparison, define

$$e_\eta(t) = \frac{\|\eta_\theta(t) - \eta_{CS}(t)\|_{L^2}}{\|\eta_{CS}(t)\|_{L^2}}, \quad e_\xi(t) = \frac{\|\xi_\theta(t) - \xi_{CS}(t)\|_{L^2}}{\|\xi_{CS}(t)\|_{L^2}}. \quad (30)$$

Potential fields are compared in the mean-zero gauge. For saved times $t_0, \dots, t_{N_t}$, the surface trajectory error is

$$E_{\eta, \text{traj}} = \left(\frac{\sum_{n=0}^{N_t} \|\eta_\theta(t_n) - \eta_{CS}(t_n)\|_{L^2}^2}{\sum_{n=0}^{N_t} \|\eta_{CS}(t_n)\|_{L^2}^2} \right)^{1/2}, \quad (31)$$

with $E_{\xi, \text{traj}}$ defined analogously. The terminal errors are $e_\eta(T)$ and $e_\xi(T)$.

4.3.1 Overall Trajectory Accuracy and Stability

Figure 3 shows the historical C27 errors over the 285 reference-valid cases. Because the horizons differ between families, the horizontal coordinate is normalized time t/T , rather than a common physical time. The pooled median terminal errors are $e_\eta(T) = 2.63 \times 10^{-3}$ and $e_\xi(T) = 2.17 \times 10^{-3}$. All 285 paired learned trajectories have finite saved states and positive saved fluid depth. This saved-state check does not establish absence of failures at unsaved internal stages. Three invalid reference cases are excluded explicitly, rather than silently counted as successful comparisons. [The final test should include a matched vanilla FNO, trajectory-disjoint initial conditions, declared internal failure criteria, terminal and trajectory error tails, and completion rates.](#)

4.3.2 Regime-Specific Physical Accuracy

Traveling Stokes and Tanaka Waves For a periodic traveling wave, define the translation error by

$$\Delta x(t) = \underset{s \in [-L/2, L/2]}{\operatorname{argmin}} \|\eta_\theta(\cdot, t) - \eta_{CS}(\cdot - s, t)\|_{L^2}. \quad (32)$$

The translation-aligned profile error is

$$e_{\eta, \text{align}}(t) = \frac{\|\eta_\theta(\cdot, t) - \eta_{CS}(\cdot - \Delta x(t), t)\|_{L^2}}{\|\eta_{CS}(\cdot, t)\|_{L^2}}. \quad (33)$$

Figure 4 separates phase drift from profile distortion for a Stokes and a single-crest Tanaka case. Representatives are selected at the median terminal surface error within the corresponding historical profile class. Alignment is computed by continuous Fourier interpolation. For Stokes, equivalent minimizers are resolved modulo one carrier wavelength and the displacement is unwrapped in time. Both ordinary and aligned errors are shown; alignment is not used to hide raw trajectory error.

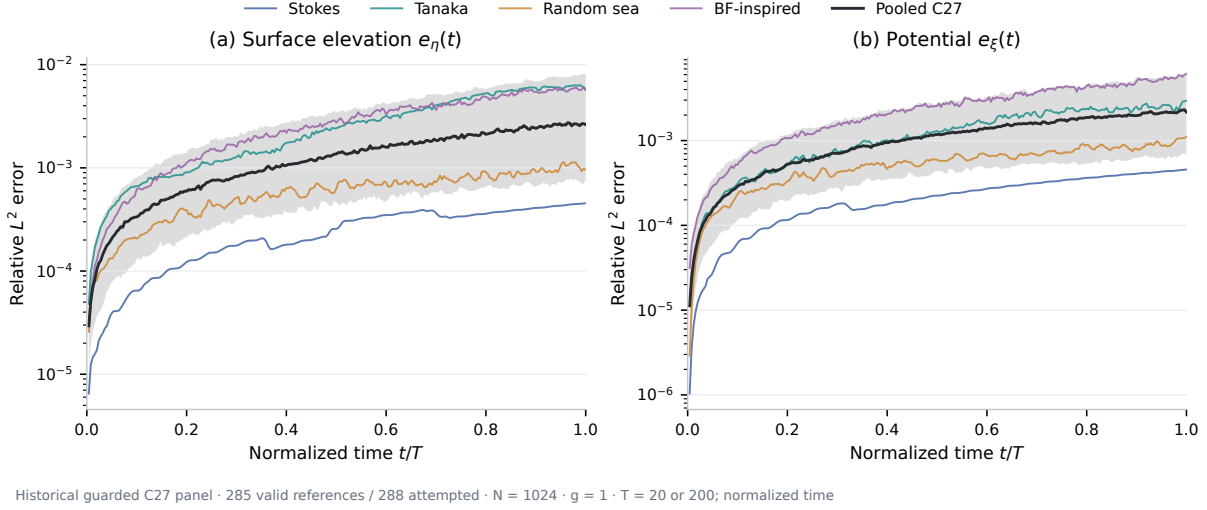


Figure 3: Relative surface and potential errors for historical guarded C27 rollouts. Black curves and shaded bands give the case-pooled median and interquartile range; colored curves give family medians. There are 285 valid reference pairs out of 288 attempted cases. The horizons are $T = 20$ for Stokes/random-sea and $T = 200$ for Tanaka/BF-inspired cases, hence the use of normalized time t/T .

Tanaka-Wave Interactions For the intended collision study, define the collision time as the time of minimum periodic separation between two tracked crests, and set

$$e_{\text{coll}} = \frac{|T_{\text{coll},\theta} - T_{\text{coll,CS}}|}{T_{\text{coll,CS}}}. \quad (34)$$

The current illustration in Figure 5 instead selects an interaction event from a local maximum of reference crest amplification. It compares the profiles before, near, and after this event. A single fixed spatial recentering is applied to both methods at all three times; there is no separate alignment of predicted profiles. [Tracked-crest collision times and the associated statistic in \(34\) have not yet been computed.](#)

Benjamin-Feir and JONSWAP/TMA Waves For periodic wavenumbers $k = 2\pi n/L$, define

$$\lambda_k(h) = |k| \tanh(h|k|), \quad \mathcal{E}_k(t) = \frac{1}{2} \left[g|\widehat{\eta}_k(t)|^2 + \lambda_k(h)|\widehat{\xi}_k(t)|^2 \right]. \quad (35)$$

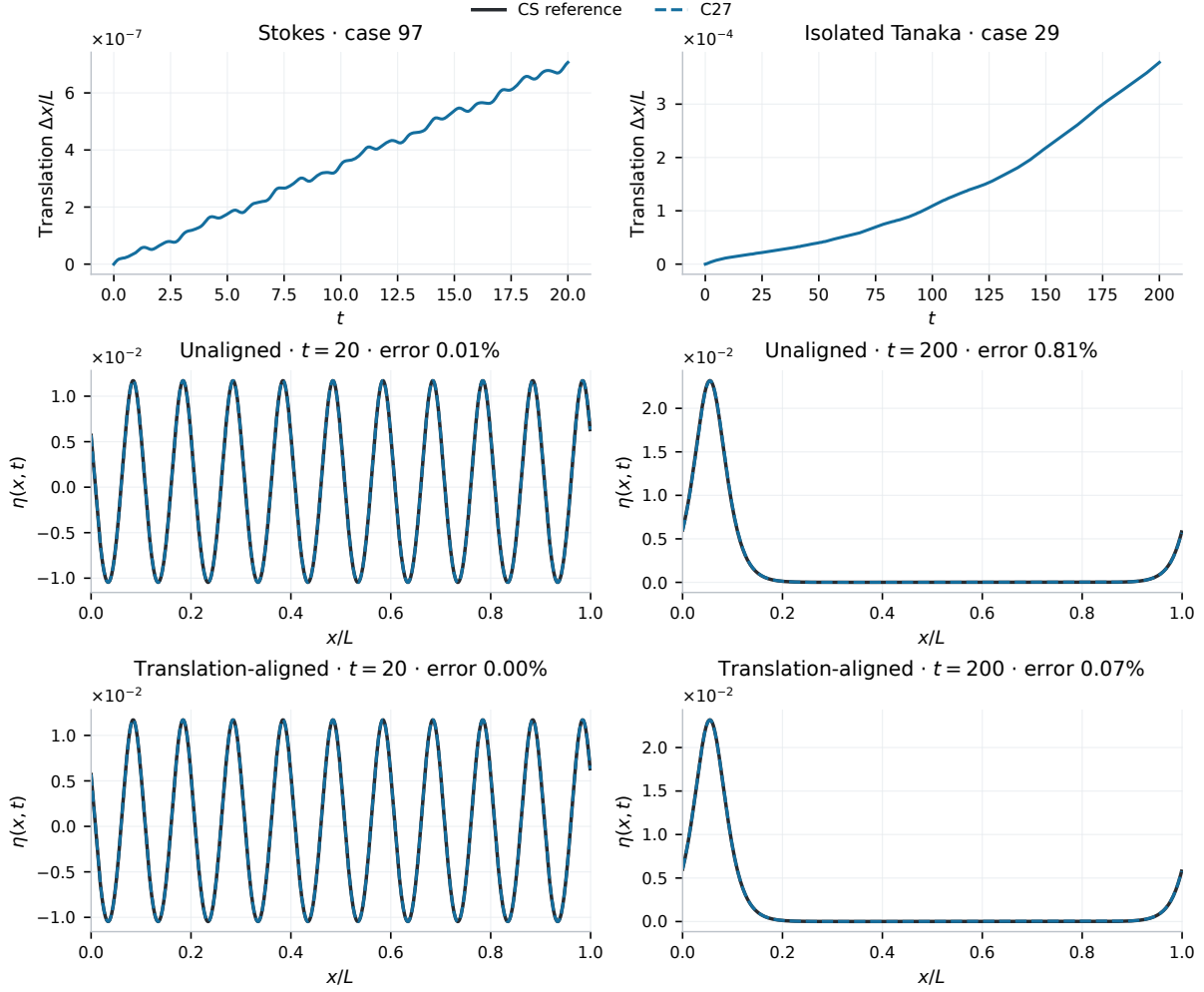
The normalized spectral-energy distribution and its transfer from the initial state are

$$p_k(t) = \frac{\mathcal{E}_k(t)}{\sum_j \mathcal{E}_j(t)}, \quad \Delta p_k(t) = p_k(t) - p_k(0). \quad (36)$$

For a carrier k_0 with symmetric sidebands spaced by q , define

$$R_{\text{sb}}(t) = \frac{\mathcal{E}_{k_0-q}(t) + \mathcal{E}_{k_0+q}(t)}{\mathcal{E}_{k_0}(t)}. \quad (37)$$

Figure 6 supplies a preliminary illustration using the available BF-inspired and random-sea families. In the BF-inspired case, the carrier and strongest symmetric seeded sideband pair are identified from the initial energy spectrum. In the random-sea case, band-energy fractions use the fixed initial peak mode n_p , with low modes $0 < n < 0.5n_p$, peak modes $0.5n_p \leq n \leq 1.5n_p$, and high modes $n > 1.5n_p$. The one-sided spectra include conjugate-mode multiplicities. The displayed comparison is not a substitute for the intended canonical Benjamin-Feir and JONSWAP/TMA experiments.



Top: continuous Fourier alignment · middle: raw prediction · bottom: aligned prediction · historical guarded rollouts

Figure 4: Translation error and late-time profiles for representative historical Stokes and isolated Tanaka cases. Top: displacement $\Delta x/L$. Middle: raw CS and C27 profiles. Bottom: profiles after translation alignment. The titles give the terminal unaligned and aligned relative errors. Both cases use the historical guarded protocol.

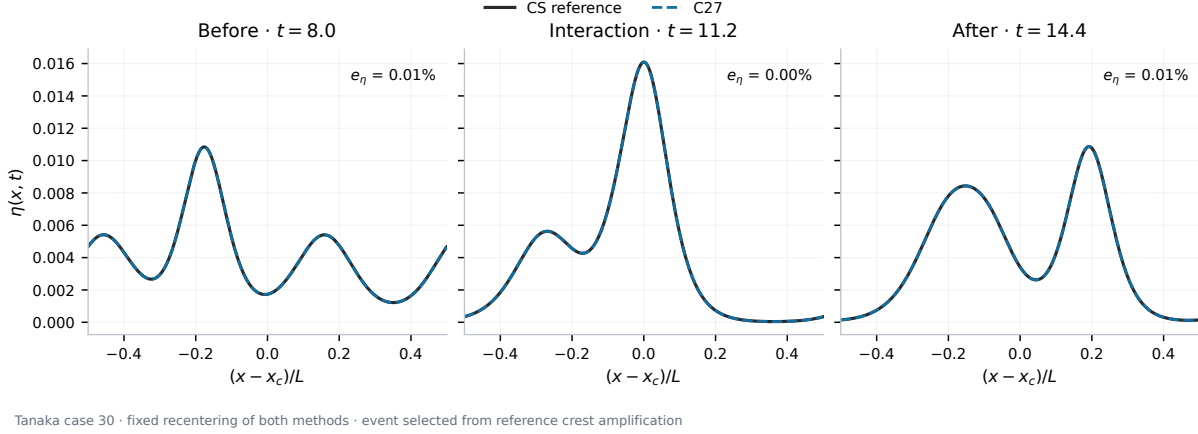


Figure 5: A representative two-crest Tanaka interaction from historical case 30, with $h = 0.125923$. The panels use identical axes and show reference and C27 profiles at $t = 8.0$, 11.2 , and 14.4 ; each reports the relative surface error. The central event is selected by reference crest amplification, not by a tracked-crest collision-time estimator.

4.4 Hamiltonian Conservation

To assess physical energy independently of the learned DNO, we evaluate the Hamiltonian of a predicted state using the reference Craig–Sulem operator:

$$H_{\text{ref}}^{\theta}(t) = \frac{1}{2} \int_{\Omega} [\xi_{\theta}(t) G(\eta_{\theta}(t); h) \xi_{\theta}(t) + g \eta_{\theta}(t)^2] dx. \quad (38)$$

The signed and absolute relative drifts are

$$\delta_H(t) = \frac{H_{\text{ref}}^{\theta}(t) - H_{\text{ref}}^{\theta}(0)}{|H_{\text{ref}}^{\theta}(0)|}, \quad e_H(t) = |\delta_H(t)|. \quad (39)$$

We freshly evaluate CS-6 with padding factor eight on both reference and predicted states at 26 saved times per trajectory. This replaces the archived learned-DNO energy metric with the physical diagnostic specified above. Figure 7 shows the signed pooled median and interquartile band, together with the representative cases used elsewhere in this draft. The median terminal absolute drift is approximately 3.60×10^{-4} for C27 states, compared with 2.09×10^{-9} for reference states. Thus these results do not place C27 drift at the reference solver floor. The numerical reference, spectral truncation, guard, and single-precision storage should all be distinguished from exact continuous Hamiltonian conservation.

4.5 M-Independent Evaluation and Computational Cost

4.5.1 FNO Complexity and Independence from M

For a Craig–Sulem expansion of order M , the cost of one DNO evaluation is

$$C_{\text{CS}}(N_x, M) = \Theta(M^2 N_x \log N_x). \quad (40)$$

For a one-dimensional FNO-type model with L_{θ} Fourier layers, channel width C , and K retained modes, a representative evaluation cost is

$$C_{\text{FNO}}(N_x) = \mathcal{O}(L_{\theta}[C N_x \log N_x + N_x C^2 + K C^2]). \quad (41)$$

When L_{θ}, C, K are fixed, the FNO evaluation has cost $\mathcal{O}(N_x \log N_x)$ and has no dependence on M . This distinction does not imply that the learned operator is faster for every resolution, truncation order, or

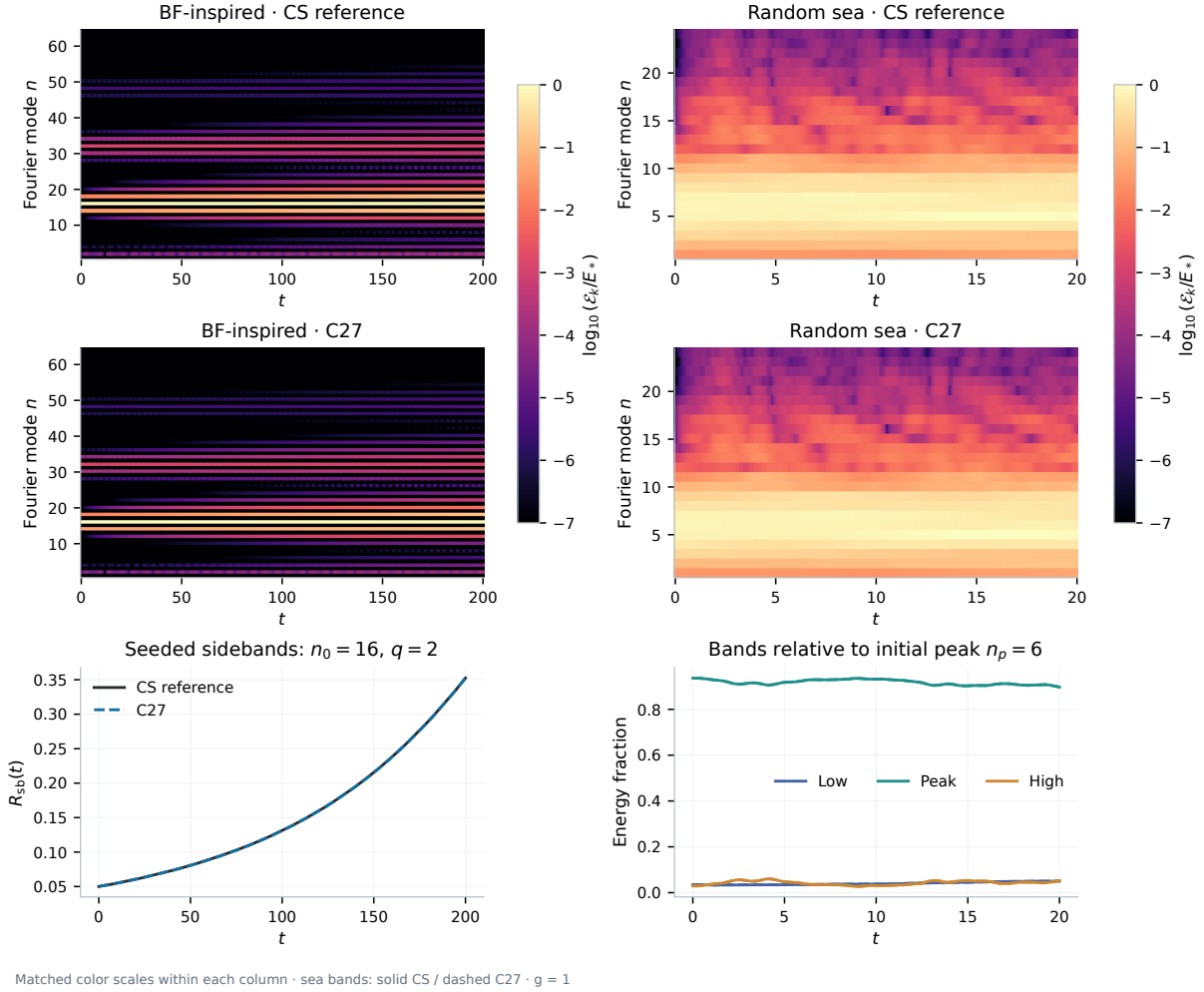


Figure 6: Quadratic spectral-energy diagnostics for historical BF-inspired and random-sea examples. The two heatmaps in each column share the same color scale, normalized by their joint maximum E_* . Below are the BF-inspired sideband-to-carrier ratio and the random-sea band fractions. For the band curves, solid lines denote CS and dashed lines denote C27. These examples use the historical guarded rollout protocol.

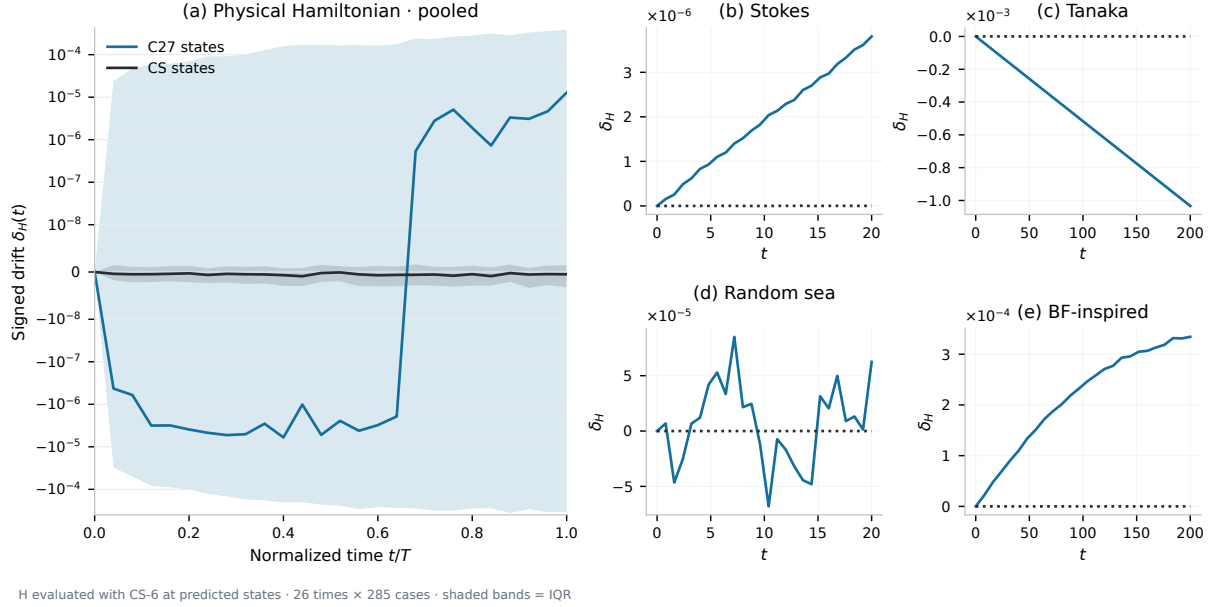


Figure 7: Relative Hamiltonian drift independently evaluated using CS-6 on both trajectories, at 26 times for each of 285 reference-valid cases. The pooled panel shows signed medians and interquartile bands on a symmetric-log scale, with linear threshold 10^{-8} . The four representative panels use linear signed axes. The C27 drift is visibly larger than the reference numerical floor.

hardware configuration. The number of DNO calls in a trajectory also depends on the time integrator and its nonlinear iteration count. [Specialize the architectural complexity expression to the final branch implementation before submission.](#)

Figure 8 measures batch-one DNO evaluation latency for $N_x \in \{128, 256, 512, 1024, 2048\}$ and CS orders $M \in \{2, 4, 6, 8\}$. A historical Stokes initial condition is Fourier-resampled to each grid. All measurements use the same CPU affinity and double precision. Seven device-synchronized measurements follow JIT warm-up; compilation is excluded. C27 has no truncation-order input, but its observed CPU latency is higher than that of the tested CS orders. Fitted exponents summarize this finite resolution range rather than prove asymptotic scaling.

4.5.2 Fixed-Horizon Trajectory Runtime

For a separate, short-horizon timing illustration we fix one historical Stokes initial condition, resample it to $N_x = 256$, and integrate to $T = 1$ with $\Delta t = 0.01$ and a hard cutoff at mode 64. Each method uses the same GL2 integration path, with four fixed-point iterations, padding factor eight, and 51 saved states. CS DNO actions and C27 are passed through the same integrator interface. Three synchronized repetitions follow warm-up, and the median wall-clock time is reported.

Figure 9 compares runtimes against CS order and surface trajectory error against runtime, with CS-10 as the numerical reference. The C27 median is 5.72 seconds, while CS orders 2, 4, 6, and 8 take 0.204, 0.480, 0.713, and 1.059 seconds, respectively. The C27 surface trajectory error is approximately 9.67×10^{-6} in this short benchmark. These CPU measurements are separate from the long-time historical panels and do not establish a wall-clock advantage for C27.

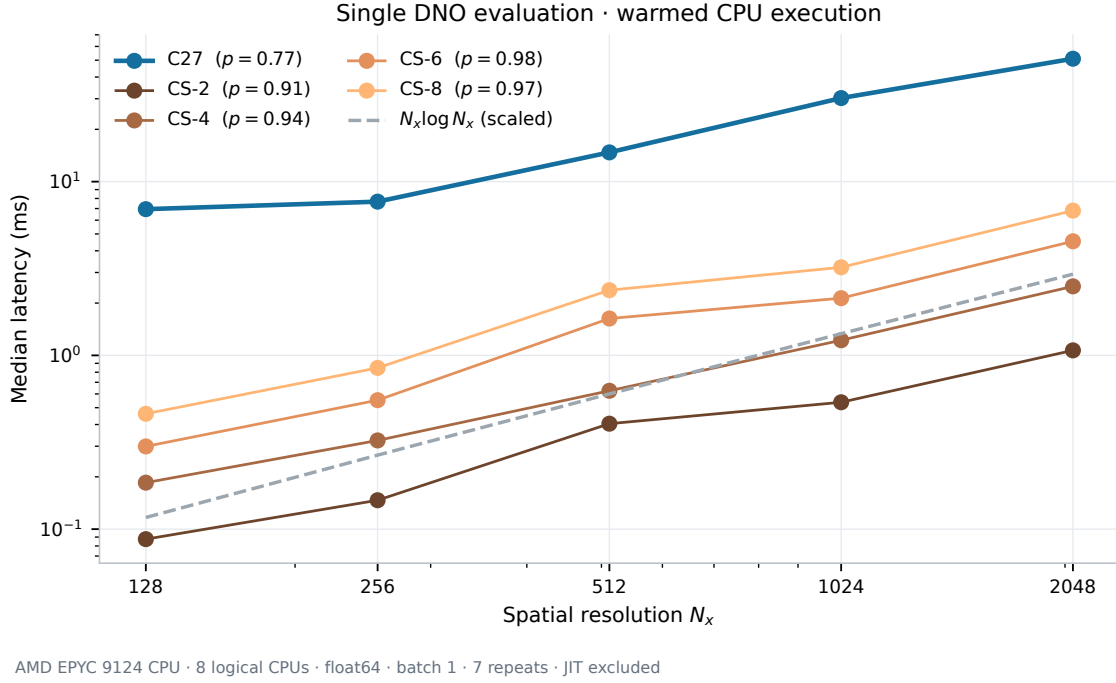


Figure 8: Warmed CPU latency for a single DNO evaluation, with batch size one and double precision. Curves show median times from seven repeats for C27 and CS orders 2, 4, 6, and 8. The legend gives finite-range log-log slopes; a scaled $N_x \log N_x$ curve is included for reference. C27 is independent of M but is slower on this CPU benchmark.

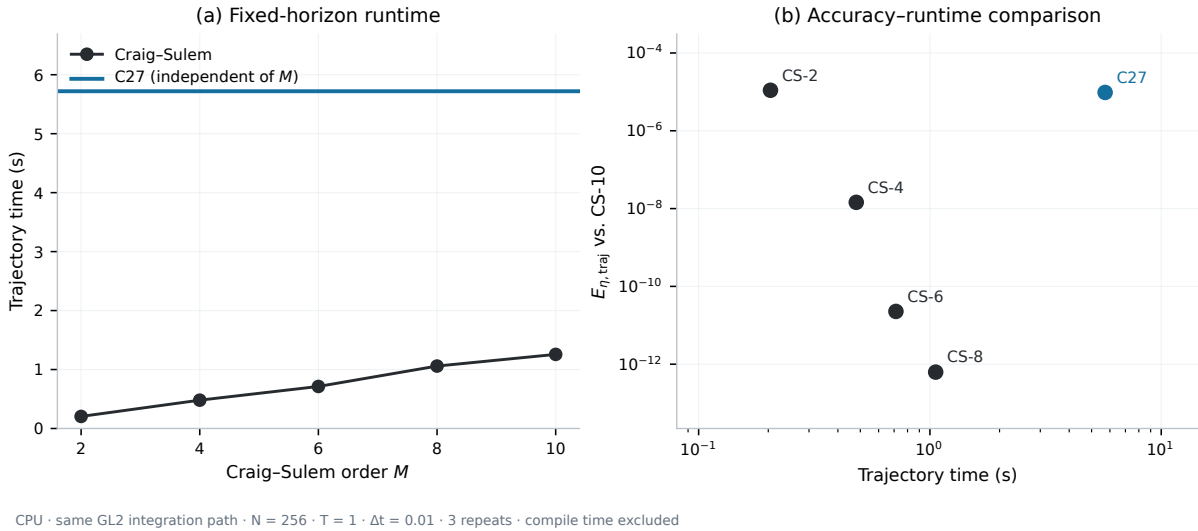


Figure 9: Short-horizon CPU runtime and accuracy for one Stokes initial condition, with $N_x = 256$, $T = 1$, and $\Delta t = 0.01$. Left: median trajectory time for CS orders 2 through 10 and the M -independent C27 time. Right: surface trajectory error against CS-10 versus runtime; the zero CS-10 self-error is omitted from the logarithmic plot. All methods use the same integration path and three warmed timing repeats.

5 Conclusion and Discussion

Limitations Complete after the final evaluation. The current figures use a sampled row split, historical guarded trajectory panels, no matched vanilla-FNO baseline, and CPU-only timing.

Future Work

Acknowledgments

This work was supported in part by...

References

Use the project's `references.bib` when the citations are completed. No bibliography file is needed to view this draft.

A Time-stepping Modal Error

We demonstrate that small errors in the Fourier modes of a learned neural surrogate for the DNO G cause long-time accumulation of modal phase error under our symplectic integration scheme.

B Low-Rankedness of Nonlinear Residuals of CS

C More About the Dataset